In [ ]:
```
# Task 2: Data Preprocessing

## Objective
To clean and prepare the dataset for machine learning.

## Steps Performed
- Checked missing values
- Removed duplicates
- Encoded categorical data
- Applied feature scaling

## Conclusion
The dataset is now clean and ready for modeling.
```

In [1]:
```
import pandas as pd
import seaborn as sns
```

In [2]:
```
df = sns.load_dataset('iris')
df.head()
```

Out[2]:

	sepal_length	sepal_width	petal_length	petal_width	species
0	5.1	3.5	1.4	0.2	setosa
1	4.9	3.0	1.4	0.2	setosa
2	4.7	3.2	1.3	0.2	setosa
3	4.6	3.1	1.5	0.2	setosa
4	5.0	3.6	1.4	0.2	setosa

In [3]:
```
df.isnull().sum()
```

Out[3]:
```
sepal_length    0
sepal_width     0
petal_length    0
petal_width     0
species         0
dtype: int64
```

In [6]:
```
df.drop_duplicates(inplace=True)
```

In [7]:
```
df.dtypes
```

Out[7]:
```
sepal_length    float64
sepal_width     float64
petal_length    float64
petal_width     float64
species          object
dtype: object
```

In [8]:
```
df['species'] = df['species'].astype('category').cat.codes
df.head()
```

Out[8]:

	sepal_length	sepal_width	petal_length	petal_width	species
0	5.1	3.5	1.4	0.2	0
1	4.9	3.0	1.4	0.2	0
2	4.7	3.2	1.3	0.2	0
3	4.6	3.1	1.5	0.2	0
4	5.0	3.6	1.4	0.2	0

In [9]:
```
from sklearn.preprocessing import StandardScaler

scaler = StandardScaler()
df_scaled = scaler.fit_transform(df)

df_scaled = pd.DataFrame(df_scaled, columns=df.columns)
df_scaled.head()
```

Out[9]:

	sepal_length	sepal_width	petal_length	petal_width	species
0	-0.898033	1.012401	-1.333255	-1.308624	-1.218613
1	-1.139562	-0.137353	-1.333255	-1.308624	-1.218613
2	-1.381091	0.322549	-1.390014	-1.308624	-1.218613
3	-1.501855	0.092598	-1.276496	-1.308624	-1.218613
4	-1.018798	1.242352	-1.333255	-1.308624	-1.218613

In [ ]: